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221764A

6th Sem/ Mech. Mech (Tool & Die Design)
Subject : Renewable Sources of Energy

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which is the largest renewable energy source in India by installed capacity?
- a) Solar
 - b) Wind
 - c) Hydro
 - d) Biomass
- Q.2 The process of converting biomass to fuel is called:
- a) Gasification
 - b) Liquefaction
 - c) Cracking
 - d) Polymerization
- Q.3 Geothermal energy comes from :
- a) Ocean tides
 - b) Earth's internal heat
 - c) Moonlight
 - d) Fossil fuel
- Q.4 Which of the following is a method of OTEC power generation?
- a) Biomass gasification
 - b) Thermal conduction
 - c) Wind turbine
 - d) Close cycle

- Q.5 Which country has one of the highest potentials for solar energy?
- a) Russia
 - b) India
 - c) UK
 - d) Canada
- Q.6 MHD stands for :
- a) Mechanical Heat Device
 - b) Magnetic High Density
 - c) Magneto Hydro Dynamics
 - d) Modern Hydro Drive

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define geothermal energy. (CO3)
- Q.8 What is the importance of solar energy? (CO2)
- Q.9 Expand OTEC. (CO3)
- Q.10 State one limitation of MHD systems. (CO4)
- Q.11 Give one environmental benefit of using biomass. (CO4)
- Q.12 Define the term "Module" in solar PV. (CO2)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Differentiate between conventional and non-conventional energy sources. (CO1)

- Q.14 Describe the construction and working principle of a photovoltaic (PV) cell. (CO2)
- Q.15 What is photosynthesis? How is it related to biomass energy generation? (CO4)
- Q.16 Explain the working of a fixed dome biogas plant. (CO4)
- Q.17 Explain the working of a wind energy conversion system with a neat sketch. (CO3)
- Q.18 Explain the Prospects of geothermal energy in India. (CO3)
- Q.19 Explain the concept of a mini and micro hydro power plant. (CO3)
- Q.20 What are the environmental impacts and benefits of biomass energy? (CO4)
- Q.21 Explain the working of a standalone solar PV system. (CO5)
- Q.22 What is the need for energy storage? Explain briefly. (CO5)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the principle and working of MHD (Magneto Hydro Dynamic) power generation system. (CO5)

- Q.24 What is a Fuel Cell? Explain the principle of working, applications advantages, and disadvantages of fuel cells. (CO3)
- Q.25 Explain the role and responsibilities of Ministry of New and Renewable Energy sources regarding free energies available in India. (CO1)